



Gujarat Technological University



NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF IT

ACADEMIC YEAR – 2025

Internship (3184201)

University ChatBot

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1. ABSTRACT

- University Chatbot is an AI-powered assistant built using Google Gemini Pro, Streamlit, and Python. It provides instant responses to university-related queries such as courses, admissions, deadlines, and more. Integrated with MongoDB and SQLite for data storage, the system aims to improve communication and support between students and institutions through natural language interaction.

2. PROBLEMS IN EXISTING SYSTEM

- Students often face delays in receiving academic information.
- Manual query resolution requires staff time and effort.
- No centralized digital assistant is available for quick support.
- Limited access to support outside working hours.

3. PURPOSE OF THE PROJECT

- To develop a chatbot that simulates a university helpdesk.
- To reduce workload on administrative staff.
- To provide 24/7 support using natural language processing.
- To deliver smart, session-based responses using Gemini API.

4. FUNCTION REQUIREMENTS

- User login/signup functionality.
- Real-time chatbot interaction.
- Session management and chat history.
- Integration with Gemini Pro for responses.
- Data storage in MongoDB and SQLite.

5. SYSTEM MODULES

- **Authentication Module:** Login & Signup
- **Chat Interface:** User input & bot reply
- **Session Handler:** Maintains conversation context
- **Database Layer:** MongoDB (cloud) & SQLite (local)
- **Gemini Integration:** For generating intelligent replies

6. SYSTEM REQUIREMENTS

➤ **Hardware:**

Minimum 4GB RAM, i3 Processor or above

➤ **Software:**

Python 3.x

Streamlit

MongoDB Atlas, SQLite

Web Browser (Chrome/Edge)

➤ **Cloud API:**

Google Gemini Pro

7. FRONT END AND BACK END OF SYSTEM

➤ **Front End:**

Developed in Streamlit

Clean UI with chat interface

➤ **Back End:**

Python for logic handling

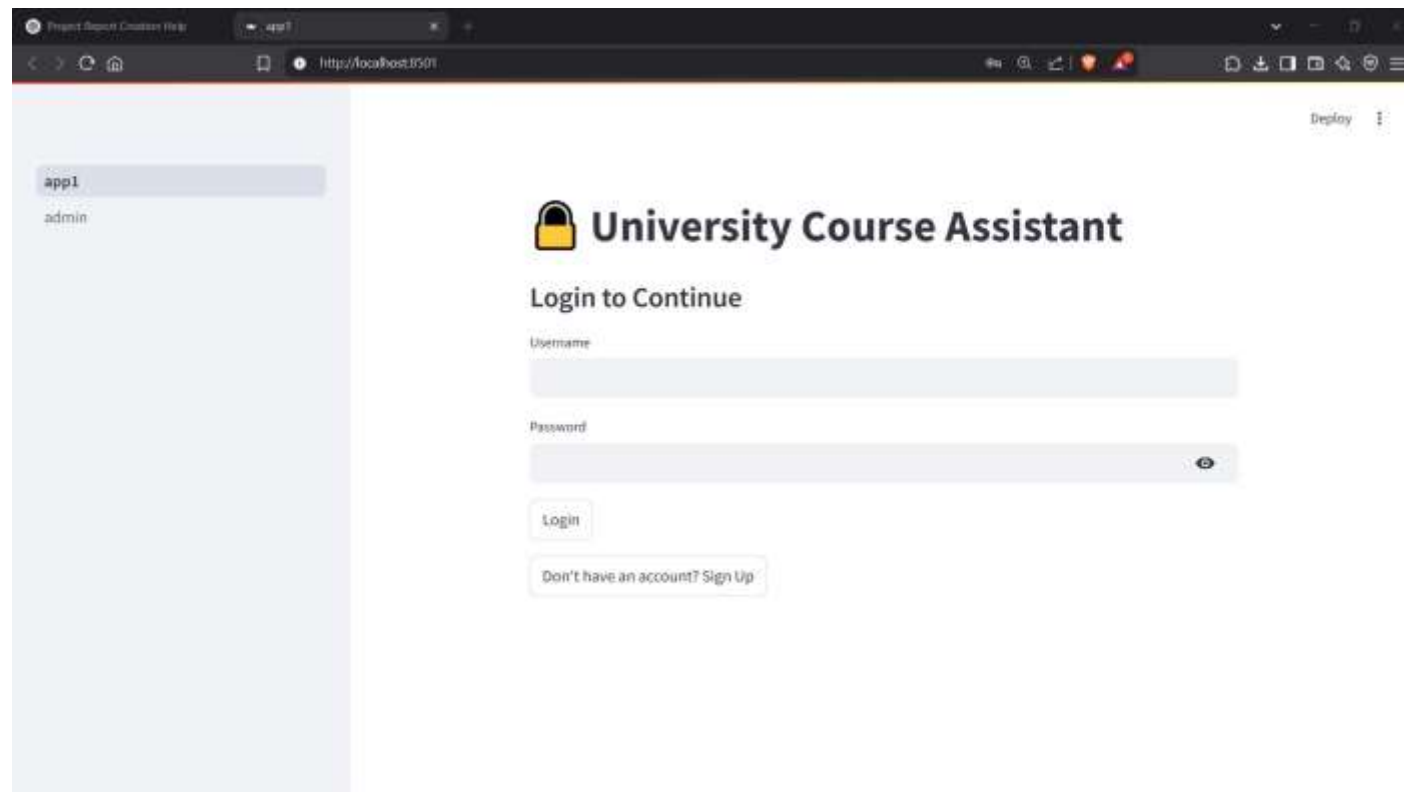
Gemini Pro for AI-based responses

MongoDB/SQLite for data management

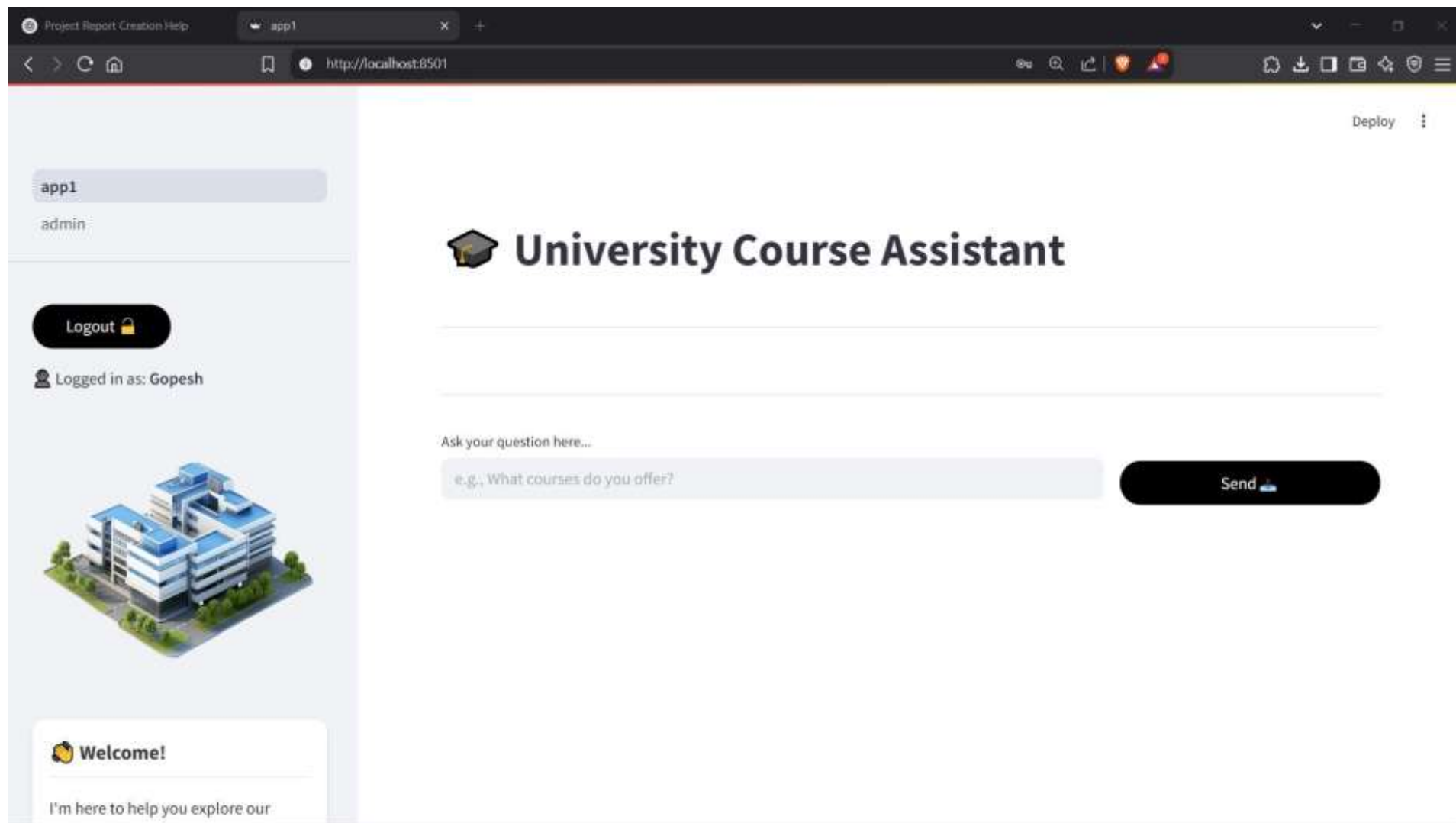
8. DATA DICTIONARY

Field Name	Description	Type
username	Registered user name	String
email	User email ID	String
message	User's question	Text
reply	Chatbot's response	Text
timestamp	Chat time and date	DateTime
session_id	Unique session ID	String

9. SNAPSHOT OF WEBSITE



Login/Sign Up page



Home Page Of chatbot

Project Report Creation Help app1 x +

http://localhost:8501

Deploy ⋮

University Course Assistant

Example Questions

- ◆ Hi! Can you help me with course information?
- ◆ What courses do you offer?
- ◆ Tell me about B.Tech program
- ◆ What is the fee structure for BCA?
- ◆ What subjects are taught in B.Sc first semester?
- ◆ How long is the B.Tech program?
- ◆ What are the subjects in BCA?
- ◆ Tell me about admission process
- ◆ What is the duration of B.Sc?
- ◆ Can you compare B.Tech and BCA programs?

You:
What subjects are taught in B.Sc first semester?

05:03

Assistant:
Hello! 😊
In the first semester of the B.Sc program, you'll be studying these subjects:

- Biology 🦋
- Chemistry 🧪
- Physics ⚡
- Mathematics ➕
- Computer Applications 💻

I hope this helps! Let me know if you have any other questions. 👍

05:03

Example of a Question

Project Report Creation Help

University Course Assistant

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http://localhost:8501/admin

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⬇️

🖨️

🔖

🛡️

☰

Deploy

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app1

admin

Navigation

- Overview
- Chat Analytics
- Course Data Management

2025/03/06

2025/04/05

Recent Conversations

📄

🔍

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	timestamp	user_message	bot_response
0	2025-04-04 12:00:40	How long is the B.Tech program?	The B.Tech program is a 4-year course, consisting of 8 semesters. 📅
1	2025-04-04 12:00:32	What is the fee structure for BCA?	The BCA program has a fee of ₹50,000 INR per semester. 💰 It's a 3-year p
2	2025-04-04 11:59:43	Hi! Can you help me with course information?	Hello! 🤖 I'd be happy to help you with information about our B.Tech, B.
3	2025-04-04 11:37:02	Hi! Can you help me with course information?	Hello! 🤖 I'd be happy to help you with information about our B.Tech, B.
4	2025-04-04 09:05:34	how are you?	I'm doing well, thank you for asking! 😊 I'm ready to assist you with any
5	2025-04-04 09:05:18	What courses do you offer?	Hello there! 🤖 I'm happy to help you with information about the course
6	2025-04-04 08:26:40	What is the fee structure for BCA?	Hello! 🤖 I'd be happy to help you with information about our courses. 1
7	2025-04-04 08:11:50	I'm Gopesh	Hello Gopesh! 🤖 It's great to meet you! How can I help you with inform
8	2025-04-04 08:06:39	what is in B.Tech?	The B.Tech program is a 4-year course divided into 8 semesters. 📅 The fr
9	2025-04-04 08:06:03	What courses do you offer?	Hello! 🤖 I can provide information about the following courses: * **B.T

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Download Chat History

Admin Panel – Chat Analysis

The image shows a Visual Studio Code editor window with a dark theme. The Explorer sidebar on the left displays a project named 'UNIVERSITY-CHATBOT' with various files including 'app1.py', 'users.json', and 'database.py'. The main editor area shows the code for 'app1.py', which includes imports for 'streamlit', 'google.generativeai', 'json', 'os', 'datetime', 'pytz', and 'database'. It defines functions for 'load_users' and 'save_user'. The bottom panel shows the 'TERMINAL' output, indicating the successful execution of 'streamlit run app1.py' and providing local and network URLs for accessing the application.

```
File Edit Selection View Go Run Terminal Help
university-chatbot

EXPLORER
UNIVERSITY-CHATBOT
  __pycache__
  .dist
  .streamlit
  pages
  Resources
  venv
  app1.py
  College-Chatbot-system-architecture...
  database.py
  DFD.png
  LICENSE
  requirements.txt
  Sequence diagram.jpg
  URL.txt
  usecase.jpg
  users.json

app1.py
1  import streamlit as st
2  import google.generativeai as genai
3  import json
4  import os
5  from datetime import datetime
6  import pytz
7  from database import init_database, get_course_data, save_chat, get_or_create_user_session
8
9  # ----- User Authentication -----
10 USERS_FILE = "users.json"
11
12 def load_users():
13     if not os.path.exists(USERS_FILE):
14         with open(USERS_FILE, "w") as f:
15             json.dump({}, f)
16     with open(USERS_FILE, "r") as f:
17         return json.load(f)
18
19 def save_user(username, password):
20     users = load_users()
21     users[username] = password
22     with open(USERS_FILE, "w") as f:
23         json.dump(users, f)
24
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\rabad\Downloads\university-chatbot> venv/scripts/activate
(venv) PS C:\Users\rabad\Downloads\university-chatbot> streamlit run app1.py

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://192.168.1.18:8501

Ln 224, Col 42 Spaces: 4 UTF-8 CRLF {} Python 3.11.0 (venv)
```

How To Run A Server

10. FUTURE ENHANCEMENT

- Add voice-based chat capabilities
- Introduce multilingual support
- Create an admin analytics dashboard
- Integrate with live university data (e.g., timetable, fees)
- Deploy with advanced user authentication

